

Social Media Performance Dashboard

Project Overview

The Social Media Performance Dashboard is an end-to-end Data Analytics project built using SQL and Power BI. The project focuses on analyzing social media performance metrics such as likes, views, comments, shares, engagement rate, followers, viral content, regional performance, and content type effectiveness.

This project demonstrates complete analytics workflow including:

- Data Cleaning
- Data Transformation
- SQL Analysis
- Data Modeling
- DAX Measures
- KPI Reporting
- Dashboard Design
- Business Insights
- Interactive Reporting

The dashboard was designed as a portfolio-ready business intelligence solution suitable for Data Analyst and BI Developer roles.

Tools & Technologies Used

Tool	Purpose
SQL (MySQL)	Data Cleaning, Transformation & Analysis
Power BI	Dashboard Development & Data Visualization
Power Query	Data Transformation
DAX	KPI Measures & Calculations
Excel/CSV Dataset	Source Data

Dataset Details

The dataset contains social media performance data from multiple platforms:

- Instagram

- Facebook
- LinkedIn
- YouTube

Dataset Columns

- POST_ID
- PLATFORM
- FOLLOWERS
- CONTENT_TYPE
- TOPIC
- LANGUAGE
- REGION
- POST_DATETIME
- POST_DATE
- POST_TIME
- HASHTAGS
- SENTIMENT_SCORE
- LIKES
- VIEWS
- COMMENTS
- SHARES
- ENGAGEMENT_RATE
- IS_VIRAL
- ENGAGEMENT_LEVEL
- FOLLOWER_CATEGORY

SQL Workflow

1. Database Creation

Created a dedicated SQL database:

```
CREATE DATABASE SOCIAL_MEDIA_PERFORMANCE_DASHBOARD;
```

2. Data Exploration

Used SQL queries to:

- View complete dataset
- Check table structure
- Validate data types

- Explore records

Example:

```
SELECT *  
FROM social_media_performance_dataset;
```

3. Datetime Conversion

Converted text datetime values into SQL DATETIME format.

```
UPDATE social_media_performance_dataset  
SET post_datetime = STR_TO_DATE(post_datetime, '%m/%d/%Y %H:%i');
```

Created separate:

- POST_DATE
- POST_TIME

columns for better analytics.

4. Data Cleaning

Performed:

- Null value checking
- Blank value checking
- Duplicate detection
- Invalid data validation
- Text trimming using TRIM()

Example:

```
SELECT *  
FROM social_media_performance_dataset  
WHERE platform IS NULL;
```

5. Feature Engineering

Created new analytical columns:

Followers Column

```
ALTER TABLE social_media_performance_dataset  
ADD followers INT;
```

Engagement Level

```
CASE  
  WHEN engagement_rate >= 0.50 THEN 'HIGH'  
  WHEN engagement_rate >= 0.20 THEN 'MEDIUM'  
  ELSE 'LOW'  
END
```

Follower Category

- INFLUENCER
 - GROWING
 - SMALL
-

6. SQL Analysis Performed

Platform Analysis

- Total posts by platform
- Average engagement by platform
- Total likes by platform
- Total shares by platform

Viral Content Analysis

- Viral posts by platform
- Viral posts by region

Content Performance Analysis

- Best content type by engagement
- Content type performance metrics

Time-Based Analysis

- Best posting hour
- Posts by month
- Best day by engagement

Advanced SQL Concepts Used

HAVING Clause

```
HAVING AVG(likes) > 5000
```

Subqueries

```
WHERE likes > (  
    SELECT AVG(likes)  
    FROM social_media_performance_dataset  
)
```

Window Functions

- ROW_NUMBER()
- RANK()
- DENSE_RANK()
- Running Totals

Example:

```
RANK() OVER (  
    ORDER BY likes DESC  
)
```

Joins

Implemented:

- INNER JOIN
- LEFT JOIN

using a separate creators table.

Power BI Workflow

1. Data Import

Imported cleaned dataset into Power BI Desktop.

2. Power Query Transformation

Performed:

- Data type correction
 - Column renaming
 - Date formatting
 - Time formatting
 - Data cleaning
 - Column arrangement
-

3. DAX Measures Created

Total Likes

```
Total Likes = SUM('social_media_performance'[LIKES])
```

Total Views

```
Total Views = SUM('social_media_performance'[VIEWS])
```

Total Comments

```
Total Comments = SUM('social_media_performance'[COMMENTS])
```

Total Shares

```
Total Shares = SUM('social_media_performance'[SHARES])
```

Average Engagement

```
Average Engagement = AVERAGE('social_media_performance'[ENGAGEMENT_RATE])
```

Viral Posts

```
Viral Posts = CALCULATE(COUNTROWS('social_media_performance'),  
'social_media_performance'[IS_VIRAL] = 1)
```

Total Followers

```
Total Followers =  
SUMX(  
    VALUES('social_media_performance'[PLATFORM]),  
    MAX('social_media_performance'[FOLLOWERS])  
)
```

Dashboard Pages

1. Dashboard Overview

Contains:

- KPI Cards
- Platform Performance Charts
- Monthly Trend Analysis
- Region Performance Map
- Content Type Distribution
- Interactive Slicers

2. Platform Analysis

Analyzed:

- Likes by Platform
 - Views by Platform
 - Engagement by Platform
 - Followers by Platform
-

3. Trend Analysis

Visualized:

- Monthly Views Trend
 - Engagement Trend
 - Posting Activity Over Time
-

4. Region Insights

Included:

- Region Performance Map
 - Regional Engagement Comparison
 - Viral Content Distribution
-

5. Content Insights

Analyzed:

- Content Type Performance
 - Viral Content Distribution
 - Topic Analysis
 - Sentiment Insights
-

Dashboard Features

KPI Cards

The dashboard includes:

- Total Followers
 - Total Likes
 - Total Views
 - Total Comments
 - Total Shares
 - Average Engagement Rate
 - Viral Posts
-

Interactive Slicers

Implemented slicers for:

- Platform
 - Region
 - Topic
 - Viral Status
 - Date
-

Data Visualizations Used

- KPI Cards
 - Bar Charts
 - Line Charts
 - Pie Charts
 - Donut Charts
 - Maps
 - Slicers
-

Business Insights Generated

- YouTube generated highest views across platforms.
 - Instagram showed strong engagement performance.
 - Certain regions consistently produced viral content.
 - High-engagement posts were associated with specific content types.
 - Posting time significantly impacted engagement rate.
-

Skills Demonstrated

SQL Skills

- Database creation
- Data cleaning
- Data transformation
- Aggregate functions
- GROUP BY & HAVING
- Subqueries
- Window functions
- Joins
- Analytical queries

Power BI Skills

- Data modeling
- Power Query
- DAX calculations
- KPI development
- Dashboard design
- Interactive filtering
- Data storytelling
- Business intelligence reporting

Project Outcome

Successfully developed a professional multi-page Social Media Performance Dashboard capable of:

- Tracking social media KPIs
- Monitoring engagement trends
- Comparing platform performance
- Identifying viral content
- Supporting business decision-making

The project demonstrates practical Data Analytics workflow from raw data to business intelligence dashboard.

GitHub Repository Structure

```
Social-Media-Performance-Dashboard/  
|  
├─ README.md  
├─ Social_Media_Performance_Dashboard.pbix  
├─ PowerBI_Transformed_social_media_performance_dataset.csv  
├─ Social media performance Queries.sql  
├─ screenshots/  
|   └─ dashboard_overview.png  
|   └─ trend_analysis.png  
|   └─ region_analysis.png
```

LinkedIn Project Description

 Built a complete Social Media Performance Dashboard using SQL and Power BI.

Key Highlights:

- Performed data cleaning and transformation using SQL and Power Query
- Built DAX measures for KPI analysis
- Created interactive Power BI dashboards with slicers and maps
- Conducted platform, regional, engagement, and content analysis
- Used advanced SQL concepts including Joins, Subqueries, HAVING Clause, and Window Functions

Tools Used:

- SQL
- Power BI
- DAX
- Power Query
- Excel/CSV

PowerBI #SQL #DataAnalytics
#BusinessIntelligence #DataAnalyst #Dashboard

Conclusion

This project demonstrates end-to-end Business Intelligence and Data Analytics capabilities using SQL and Power BI. It highlights practical skills in data cleaning, transformation, analysis, visualization, and dashboard development while delivering meaningful business insights from social media performance data.